

Unit One Assignment

Configure and Administer Server

Complete Answer Sheet

Sections 1.1 · 1.2 · 1.3 · 1.4

Subject: Server Administration

Unit: Unit One

Total Questions: 20

1.1

Overview of Server Operating System

Q1. What is a server operating system?

Answer: A server operating system (Server OS) is a type of operating system specifically designed to run on a server computer and manage network resources. It provides services such as file sharing, user authentication, and application hosting to multiple client computers connected over a network. Examples include Windows Server 2019 and Ubuntu Server.

Q2. State two functions of a server OS.

Answer: **1. User Management and Authentication:** The server OS manages user accounts, controls who can log in to the network, and verifies user credentials before granting access to resources. **2. Resource Sharing and Management:** It allows multiple users to share hardware resources such as files, printers, and internet connections from a central location.

Q3. Give two examples of server operating systems.

Answer: **1. Windows Server 2019 / 2022** — Microsoft's enterprise server OS with Active Directory, Hyper-V, and IIS built in. **2. Ubuntu Server (Linux)** — A free, open-source server OS widely used for web hosting, cloud computing, and file servers.

Q4. What is the difference between a desktop OS and a server OS?

Answer: A **desktop OS** (such as Windows 11) is designed for a single user performing everyday tasks like browsing, gaming, and document editing. It has a full graphical interface and limited network management features. A **server OS** (such as Windows Server 2022) is designed to serve hundreds or thousands of users simultaneously over a network. It focuses on stability, security, centralised management, and providing services — it often runs without a full GUI to save resources. Server OS also supports features like Active Directory, DHCP, and DNS which a desktop OS does not.

Q5. Why is security important in a server operating system?

Answer: Security is critical in a server OS because the server stores sensitive data such as user credentials, financial records, and confidential company files. A security breach can affect every user connected to the network. The server OS enforces access control policies, protects against unauthorised access, encrypts data, applies security patches, and monitors for threats — ensuring the integrity and availability of all network resources.

1.2**Confirming Network Operating System****Q1. What is a network operating system?**

Answer: A Network Operating System (NOS) is an operating system designed to support and manage network functions such as sharing resources, managing user accounts, and controlling communication between computers on a network. It runs on a server and allows multiple client computers to connect and use shared resources such as files, printers, and internet access from one central location.

Q2. Mention two features of a network OS.

Answer: **1. Centralised User Management:** The NOS provides a single point for creating, modifying, and deleting user accounts and access permissions across the entire network. **2. Resource Sharing:** It enables network users to share hardware (printers, storage) and software resources efficiently without each computer needing its own copy or device.

Q3. Why do organisations use network operating systems?

Answer: Organisations use a network OS because it allows them to manage all computers and users from one central server rather than configuring each machine individually. This reduces cost, improves security through centralised policies, makes data backup easier, allows remote access management, and ensures all users work under the same rules and standards. It also improves collaboration since files and resources can be shared across the organisation.

Q4. Name two common network operating systems.

Answer: **1. Windows Server 2019/2022** — The most widely used NOS in enterprise environments, offering Active Directory, DNS, DHCP, and many other network services. **2. Linux (Ubuntu Server / CentOS / Red Hat Enterprise Linux)** — Open-source NOS widely used in businesses, web hosting, and cloud environments due to its stability and low cost.

Q5. What is the role of a network administrator?

Answer: A network administrator is responsible for installing, configuring, and maintaining the network and server infrastructure of an organisation. Their duties include: managing user accounts and access permissions, setting up and maintaining servers, monitoring network performance and security, troubleshooting network problems, applying software updates and security patches, configuring firewalls and backups, and ensuring network services such as internet, email, and file sharing remain available to all users.

1.3

Identifying Product, Vendor Architecture and Equipment

Q1. What is meant by server architecture?

Answer: Server architecture refers to the design and structure of a server system — including its hardware components (processor, RAM, storage, network cards), software configuration, and how these components work together to deliver services. It also includes whether the server is physical (bare metal), virtual, or cloud-based, and how it is organised within the network (e.g. client-server model, three-tier architecture). Good server architecture ensures performance, scalability, and reliability.

Q2. Name two server hardware vendors.

Answer: **1. Dell Technologies (Dell EMC)** — One of the world's largest server manufacturers, known for the PowerEdge server series used in businesses of all sizes. **2. Hewlett Packard Enterprise (HPE)** — A leading server vendor known for the ProLiant server series, widely used in enterprise and data centre environments.

Q3. What is the function of a server processor?

Answer: The server processor (CPU — Central Processing Unit) is the brain of the server. Its functions include: executing instructions from the operating system and applications, handling multiple simultaneous requests from many network users, managing data processing tasks such as calculations and database queries, coordinating communication between other hardware components (RAM, storage, network cards), and running virtualisation if the server hosts virtual machines. Server processors (such as Intel Xeon or AMD EPYC) are designed with more cores and higher performance than desktop processors.

Q4. Define computer equipment in networking.

Answer: Computer equipment in networking refers to all the physical hardware devices used to build, connect, and maintain a computer network. This includes: **servers** (central computers providing services), **switches** (connect devices within a local network), **routers** (connect different networks and direct traffic), **network cables and patch panels** (physical connections), **firewalls** (security devices controlling traffic), **access points** (provide wireless connectivity), and **client computers** (end-user machines that access network resources).

Q5. Why is vendor selection important?

Answer: Vendor selection is important because the choice of hardware and software vendor directly affects the reliability, performance, cost, and long-term support of the server infrastructure. Choosing the right vendor ensures: compatibility with existing systems, availability of technical support and warranty services, access to regular firmware and security updates, good value for the organisation's budget, and a trustworthy supply chain for future hardware upgrades. A poor vendor choice can lead to frequent failures, unsupported equipment, and high maintenance costs.

1.4

Identifying Technology and Resource

Q1. What is technology resource identification?

Answer: Technology resource identification is the process of discovering, cataloguing, and documenting all the technology assets available within an organisation's IT infrastructure. This includes identifying hardware (servers, computers, network devices), software (operating systems, applications, licences),

storage systems, network resources, and human resources (IT staff skills). It is essential for planning, budgeting, and managing the IT environment effectively.

Q2. Give examples of server resources.

Answer: **Hardware Resources:** CPU (processor cores), RAM (memory), hard drives / SSDs (storage), network interface cards (NICs), and power supply units. **Software Resources:** The server operating system, installed roles and services (DNS, DHCP, Active Directory, IIS), database software (SQL Server), and backup software. **Network Resources:** IP addresses, bandwidth, shared folders, printers, and internet connections provided to network users.

Q3. What is the importance of RAM in a server?

Answer: RAM (Random Access Memory) is critically important in a server because it is the temporary working memory where the processor stores data it is actively using. In a server environment, RAM determines how many users and applications the server can handle simultaneously. More RAM means: faster response times for users, ability to run more services at once, better performance for databases and virtual machines, and reduced need to use slower disk storage as overflow memory (paging). Insufficient RAM is one of the most common causes of poor server performance.

Q4. Define storage technology.

Answer: Storage technology refers to the hardware and software systems used to store, organise, and retrieve digital data on a server or network. Types include: **HDD (Hard Disk Drive)** — traditional magnetic storage, high capacity, lower cost; **SSD (Solid State Drive)** — faster, more reliable, no moving parts; **NAS (Network Attached Storage)** — dedicated storage device accessible over the network; **SAN (Storage Area Network)** — high-performance dedicated storage network used in enterprises; **RAID (Redundant Array of Independent Disks)** — combines multiple drives for speed or redundancy to protect against data loss.

Note: RAID is a key concept — RAID 1 mirrors data for backup, RAID 5 balances speed and redundancy.

Q5. Why should server resources be monitored?

Answer: Server resources should be monitored continuously because: **(1)** It helps detect performance problems early before they affect users — such as high CPU usage, low RAM, or full storage. **(2)** It ensures the server is available and responsive at all times (uptime). **(3)** It helps identify security threats such as unusual network traffic or unauthorised access attempts. **(4)** It provides data for capacity planning — knowing when to upgrade hardware before the server becomes overloaded. **(5)** It helps meet Service Level Agreements (SLAs) that guarantee a certain level of performance and availability to users.

Note: Windows Server tool for monitoring: Task Manager, Resource Monitor, Performance Monitor (perfmon.msc)

All 20 questions answered across Sections 1.1, 1.2, 1.3, and 1.4. Answers are written in clear academic language suitable for assignment submission.